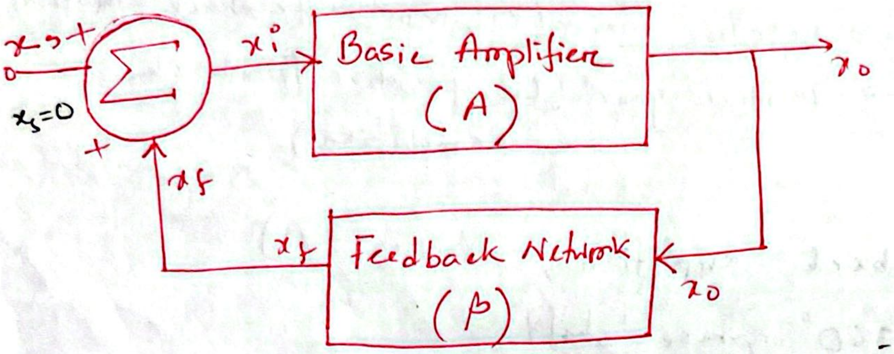
Oscillators:

- A circuit that converts a DC power source into a continuous, periodic alternating waveform at a specific frequency, without needing an external input signal.
- Signal Generator.
- Important for electrical and communication Engineering

Basic Structure:



Signal flow diagram of Positive Feedback (FB) Amplifier.

$x_i = x_s + x_f$

$= x_s + \beta x_o, \quad x_f = \beta x_o$

$= x_s + \beta A x_i, \quad x_o = A x_i$

$\therefore x_s = x_i - \beta A x_i$

$= x_i(1-\beta A)$

Gain with feedback,

$$A_f = \frac{x_o}{x_s} = \frac{A x_i}{x_i(1-\beta A)}$$

$$\therefore \boxed{A_f = \frac{A}{1-A\beta}, \quad A_f > A}$$

* Oscillator circuits work on the principle of positive FB Amplifier.

If,

$1 - A\beta = 0$

$$A_f = \frac{x_o}{x_s} = \frac{A}{1-A\beta} = \infty$$

* It is possible to get an output signal x_o even with zero input signal, i.e., $x_s = 0$

If. $1 - A\beta = 0 \Rightarrow A\beta = 1$

$A\beta \rightarrow$ Loop Gain

$x_f = \beta A x_i$. Loop gain $= \frac{x_f}{x_i} = A\beta$

$$\boxed{x_f = x_i, \text{ if } A\beta = 1}$$

Loop Gain (L) $A\beta = 1\angle 0^\circ$

$|L| = 1$

$\angle L = 0$

Barkhausen Criterion for sustained oscillation

* Feedback network ⇒ frequency Selective Circuit (β)

Basic Amplifier
- → negative gain (CE, negative feedback amplifier)
- → positive gain (CB, positive feedback amplifier)

1. For positive feedback amplifier, (Gain $+A$)
   $0^\circ/360^\circ$ phase shift.

   Frequency Selective Circuit
   $0^\circ/360^\circ$ phase shift.

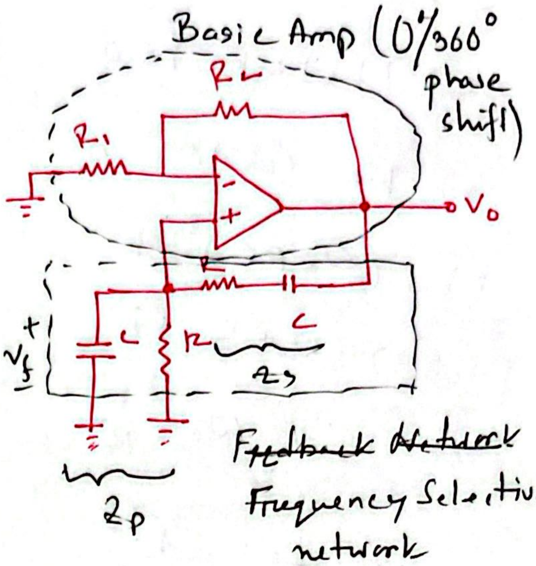
2. For negative feedback amplifier, (Gain $-A$)
   180° phase shift.

   Frequency selective Circuit.
   180° phase shift.

Wein-Bridge Oscillator: (positive gain amplifier)

→ Basic Amp - Non-inverting amplifier (0° Phase shift)

→ FB Network → should provide 0° or 360° phase shift..



Loop Gain,

$$L = A\beta = 1\angle 0^\circ$$

$$A = \frac{V_o}{V_f} = 1 + \frac{R_2}{R_1}$$

$$\beta = \frac{V_f}{V_o} = \frac{Z_p}{Z_p + Z_s}$$

$$Z_p = R \| \frac{1}{j\omega C}$$

$$Z_s = R + \frac{1}{j\omega C}$$

$$L = \left(1 + \frac{R_2}{R_1}\right) \frac{Z_p}{Z_p + Z_s}$$

$$= \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + Z_s/Z_p}$$

$$\frac{1}{Z_p} = Y_p = \frac{1}{R} + j\omega C$$

$$= \frac{1 + R_2/R_1}{1 + Z_s/Z_p}$$

$$= \frac{1 + R_2/R_1}{1 + 1 + \frac{1}{j\omega RC} + j\omega RC + 1} = \frac{1 + R_2/R_1}{3 + \frac{1}{j\omega CR} + j\omega RC}$$

Loop gain, $L = A\beta$

$$= \frac{1 + R_2/R_1}{3 + \underbrace{\frac{1}{j\omega RC} + j\omega RC}_{Im[L] = 0}} = 1\angle 0^\circ$$

$Re[L] = 1$

$Im[L] = 0$

$$\frac{1}{j\omega_o CR} + j\omega_o RC = 0$$

$$\Rightarrow 1 + (j\omega_o RC)^2 = 0$$

$$\Rightarrow 1 - (\omega_o RC)^2 = 0$$

$$\Rightarrow (\omega_o RC)^2 = 1$$

$$\Rightarrow \omega_o RC = 1$$

$$\therefore \omega_o = \frac{1}{RC}$$

$$\boxed{f_o = \frac{1}{2\pi RC}}$$

frequency of oscillation

$$Re[L] = 1$$

$$= Re\left[\frac{1 + R_2/R_1}{3 + \underbrace{\frac{1}{j\omega RC} + j\omega RC}_{=0}}\right] = 1$$

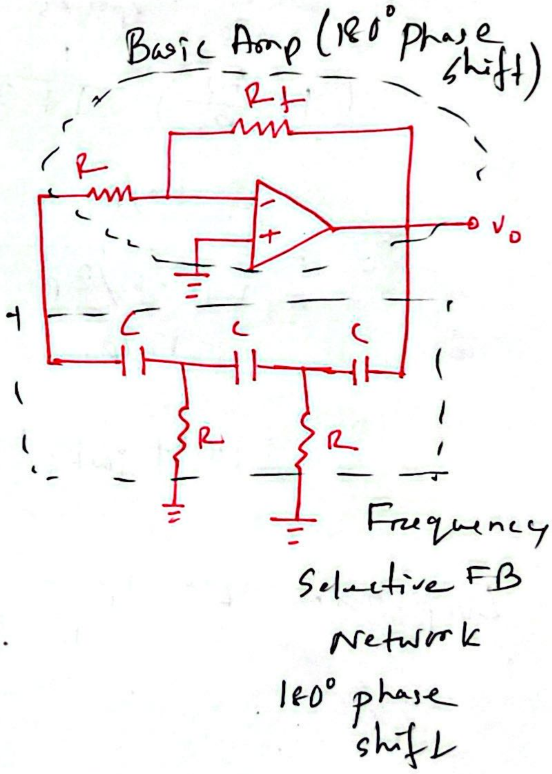
$$Re[L] = \frac{1 + R_2/R_1}{3} = 1$$

$$\Rightarrow 1 + R_2/R_1 = 3$$

$$\Rightarrow R_2/R_1 = 2$$

RC Phase Shift oscillator:

Negative Gain Amplifier.



Loop Gain,

$$L = A\beta = 1\angle 0^\circ$$

$$Re[L] = 1$$

$$Im[L] = 0$$

Frequency of oscillation,

$$f_o = \frac{1}{2\pi CR\sqrt{6}}$$

Gain of Basic Amp,

$$\frac{R_f}{R} = 29.$$